ECE 3355: Electronics

Section 14596/28468
Fall 2020

Exam 2

Version A

November 19, 2020

Complete the exam on your own, without the help of your notes, prior examples or solutions, your book, or any communication/interaction with others. You must write a complete solution that shows the relevant steps if you want full credit for the problem. You may use a calculator and a crib sheet is provided as part of this exam. **You will have 1 hr. 15 min. to finish the exam.**

Student's Name: _____

Question	Points	Score
1	30	
2	35	
3	35	
Total:	100	

From Table 6.2:

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$$

Note: For the pnp transistor, replace v_{BE} with v_{EB}.

$$i_C = \alpha i_E$$

$$i_C = \beta i_B$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$i_B = (1-\alpha)\, i_E = \frac{i_E}{\beta+1}$$

$$i_E = (\beta+1)\, i_B$$

$$\alpha = \frac{\beta}{\beta+1}$$

$V_T =$ thermal voltage $= \dfrac{kT}{q} \simeq 25\,\text{mV}$ at room temperature

Summary of Table 7.3 (Small Signal Model Parameters)

Model Parameters in Terms of DC Bias Currents

$$g_m = \frac{I_C}{V_T} \quad r_e = \frac{V_T}{I_E} = \alpha\frac{V_T}{I_C} \quad r_\pi = \frac{V_T}{I_B} = \beta\frac{V_T}{I_C} \quad r_o = \frac{|V_A|}{I_C}$$

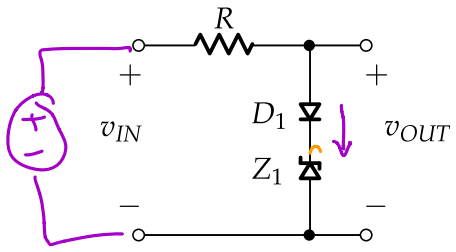
In Terms of g_m

$$r_e = \frac{\alpha}{g_m} \qquad r_\pi = \frac{\beta}{g_m}$$

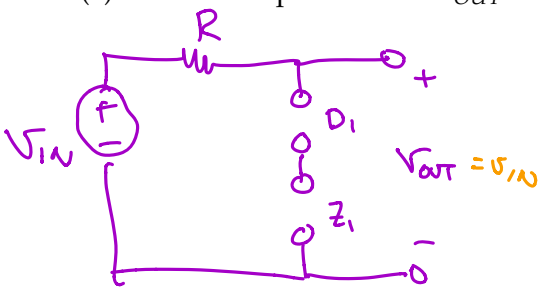
In Terms of r_e

$$g_m = \frac{\alpha}{r_e} \quad r_\pi = (\beta+1)\, r_e \qquad g_m + \frac{1}{r_\pi} = \frac{1}{r_e}$$

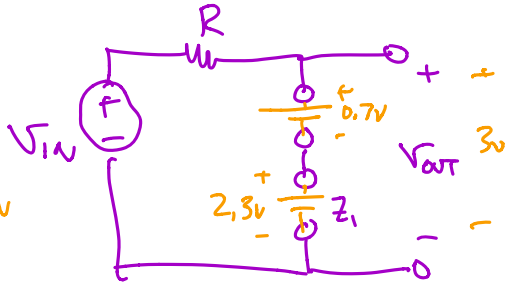
1. (30 points) The circuit below consists of a resistor (R), a regular diode (D_1), and a Zener diode (Z_1) in series. Use $V_{D0} = 0.7\,\text{V}$ (for both diodes), $V_{Z0} = 2.3\,\text{V}$, and $r_D = r_Z = 0$ to solve this problem.



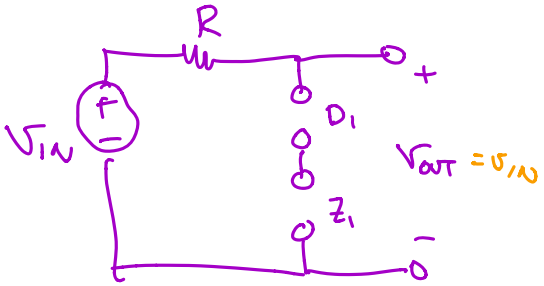
(a) Write an expression for v_{OUT} in terms of v_{IN}.

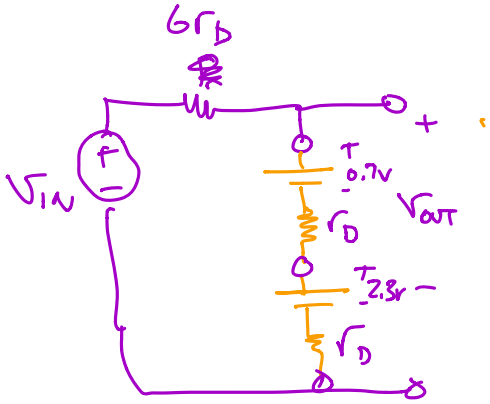


$$v_{OUT} = \begin{cases} v_{IN} & v_{IN} < 3\text{V} \\ 3\text{V} & v_{IN} \geq 3\text{V} \end{cases}$$



(b) Rewrite the expression with $r_D = r_Z$ and $R = 6r_D$.

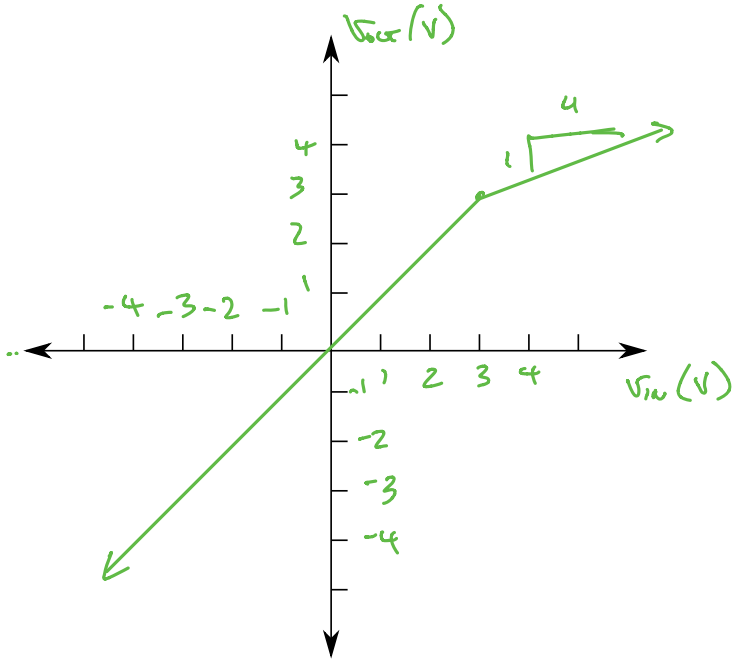




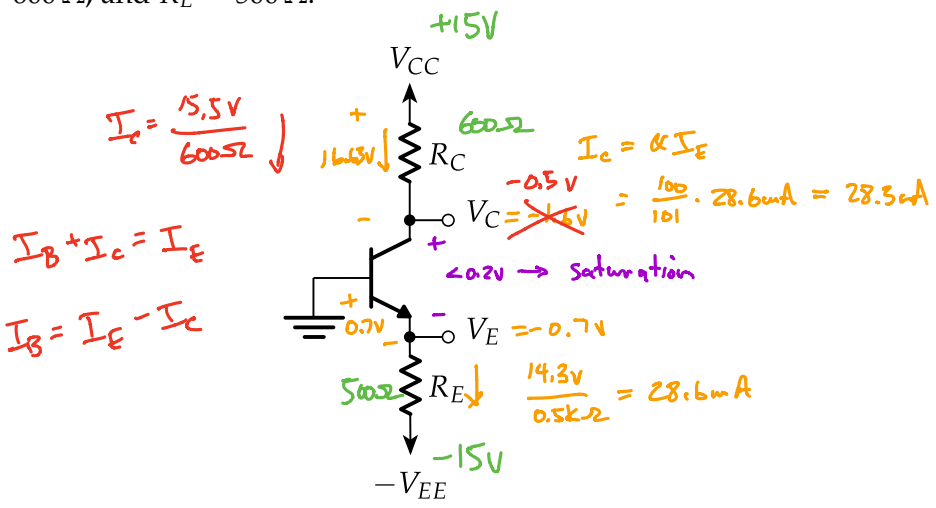
$$v_{OUT} = \begin{cases} v_{IN} & v_{IN} < 3\text{V} \\ \frac{v_{IN} - 3\text{V}}{4} \cdot \cancel{8r_D} + 3\text{V} & v_{IN} \geq 3\text{V} \end{cases}$$

$$= \begin{cases} v_{IN} & v_{IN} < 3\text{V} \\ \frac{1}{4}\left[v_{IN} - 3\text{V}\right] + 3\text{V} & v_{IN} \geq 3\text{V} \end{cases}$$

(c) Sketch a graph of v_{OUT} vs. v_{IN} on the graph provided using the expression from the second solution, Part (b).



2. (a) (15 points) Given the circuit below, determine the currents and voltages for all of the nodes of the transistor (i.e., find I_B, I_C, I_E, V_C, and V_E) given $V_{CC} = 15\,\text{V}$, $-V_{EE} = -15\,\text{V}$, $\beta = 100$, $R_C = 600\,\Omega$, and $R_E = 500\,\Omega$.



$$I_E = \frac{14.3\text{V}}{0.5\text{k}\Omega} = 28.6\text{mA}$$

$$I_c = \frac{15.5\text{V}}{0.6\text{k}\Omega} = 25.8\,\text{mA}$$

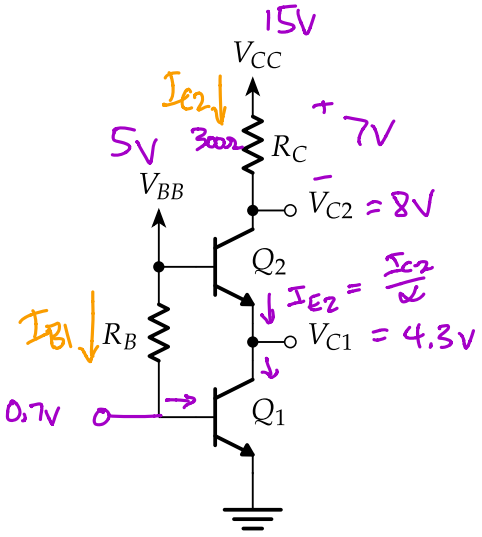
$$I_B = I_E - I_c = 2.8\,\text{mA}$$

$$V_B = 0$$

$$V_E = -0.7\text{V}$$

$$V_c = -0.5\,\text{V}$$

(b) (20 points) For the following circuit, what is the approximate value of β, which is the same for both transistors, if $V_{CC} = 15\,\text{V}$, $V_{BB} = 5\,\text{V}$, $R_B = 36.7\,\text{k}\Omega$, $R_C = 300\,\Omega$, and $V_{C2} = 8\,\text{V}$?



$I_{C2} = \frac{7V}{0.3K\Omega} = 23.\bar{3}mA$

$I_{B1} = \frac{4.3V}{36.7K\Omega} = 0.18mA$

$I_{E2} = \frac{I_{C2}}{\alpha} = I_{C1} = \beta I_{B1}$

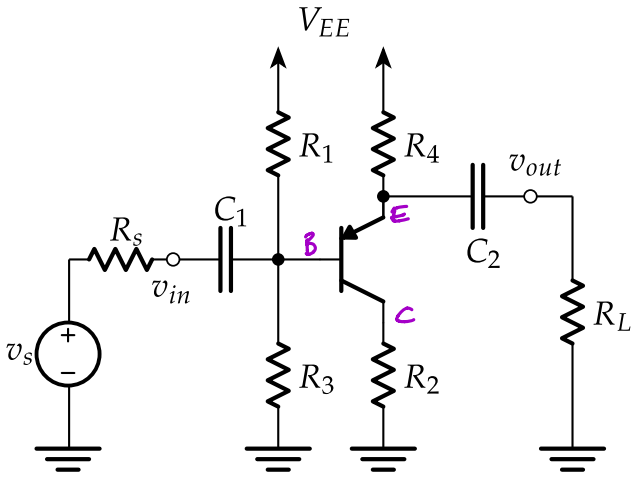
$\frac{I_{C2}}{\alpha} = \beta I_{B1}$

$\frac{I_{C2}}{I_{B1}} = \alpha \cdot \beta = \frac{23.3mA}{0.18mA} = 199$

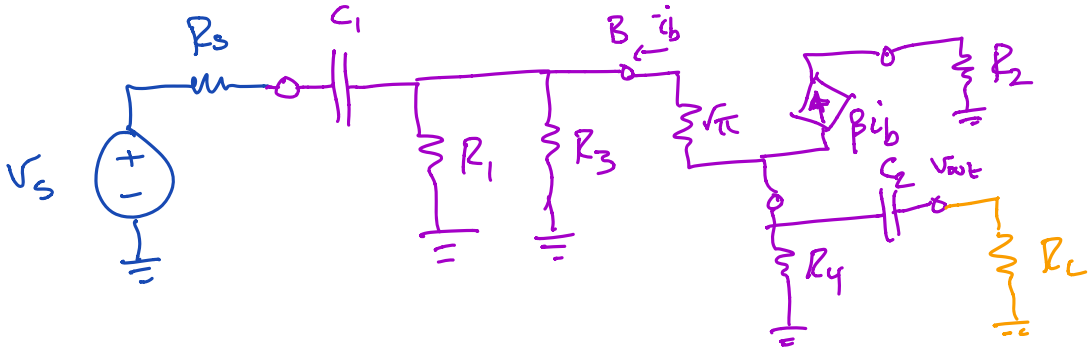
since α is a little less than 1,

$\beta \approx 200$

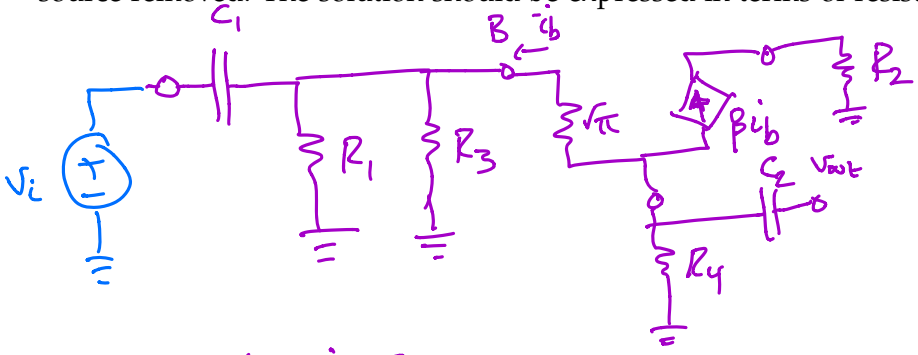
3. Use the following circuit to complete this problem.



(a) (5 points) Assuming that the DC biasing leaves the transistor in its active mode, sketch the small signal model for the circuit.



(b) (10 points) Write an expression for the *midband gain* of the circuit, $\frac{v_{out}}{v_{in}}$, with the load and source removed. The solution should be expressed in terms of resistors, β, and r_π.

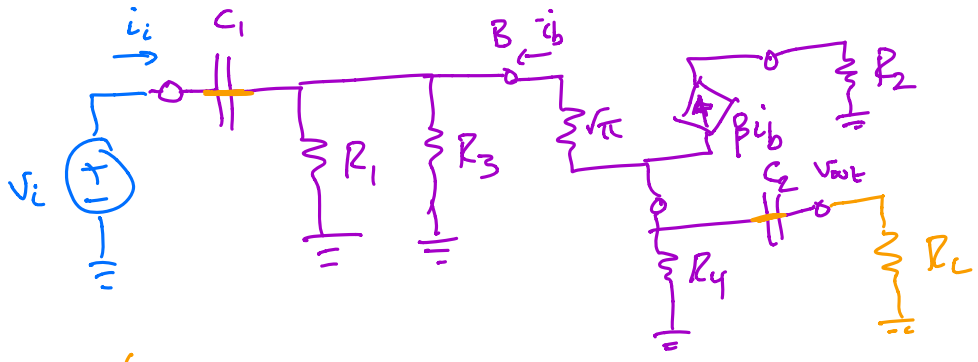


$v_{out} = -(\beta+1) i_b \cdot R_4$

$v_i = -i_b r_\pi - (\beta+1) i_b R_4$

$\frac{v_{out}}{v_{in}} = \frac{(\beta+1) R_4}{r_\pi + (\beta+1) R_4}$

(c) (10 points) Find expression for the **input resistance** for the *midband*. Note that you should find the input resistance with the load attached.



Two Approaches:

(1) $R_{in} = R_1 \| R_3 \| \dfrac{v_i}{-i_b}$

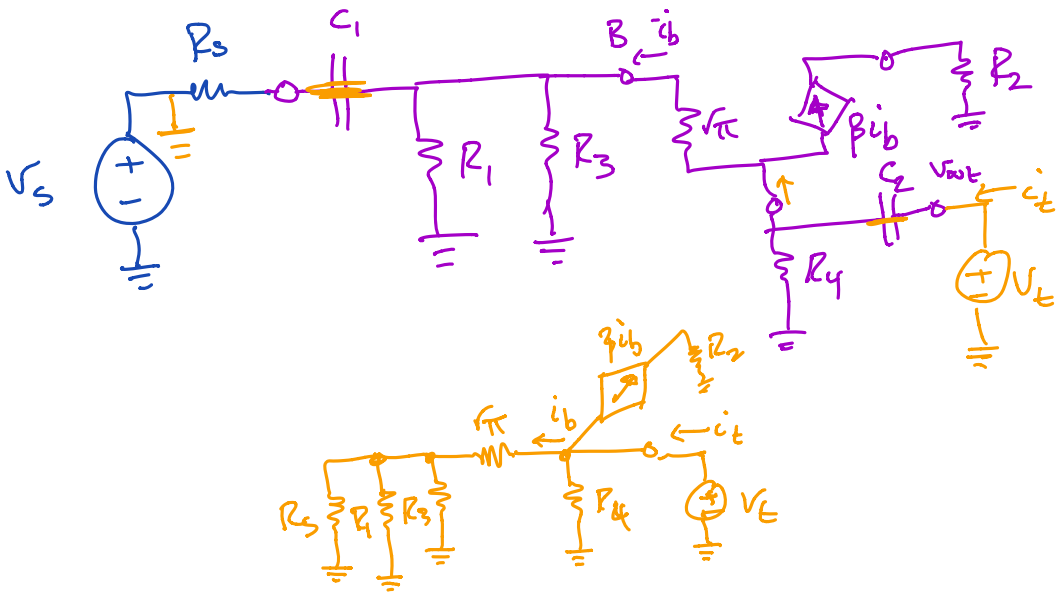
$= R_1 \| R_3 \| \dfrac{-i_b r_\pi - (\beta+1) i_b (R_4 \| R_L)}{-i_b}$

$= R_1 \| R_3 \| \left[r_\pi + (\beta+1)(R_4 \| R_L) \right]$

(2) $R_{in} = \dfrac{v_i}{i_i} \Rightarrow i_i = \dfrac{v_i}{R_1} + \dfrac{v_i}{R_3} - i_b \qquad i_b = -\dfrac{v_i}{r_\pi + (\beta+1)(R_4 \| R_L)}$

$R_{in} = \dfrac{v_i}{\frac{v_i}{R_1} + \frac{v_i}{R_3} + \frac{v_i}{r_\pi + (\beta+1)(R_4 \| R_L)}} = R_1 \| R_3 \| \left[r_\pi + (\beta+1)(R_4 \| R_L) \right]$

(d) (10 points) Find expression for the **output resistance** for the *midband*. Note that you should find the output resistance with the source attached.



Two approaches

(1) $R_o = R_4 \| \frac{V_t}{(\beta+1) i_b} = R_4 \| \left[\frac{i_b r_\pi + i_b (R_1 \| R_3 \| R_s)}{(\beta+1) i_b} \right] = R_4 \| \left[\frac{r_\pi + R_1 \| R_3 \| R_s}{\beta+1} \right]$

(2) $R_o = \frac{V_t}{i_t} \Rightarrow i_t = \frac{V_t}{R_4} + (\beta+1) i_b$ and $i_b \cdot [R_s \| R_1 \| R_3 + r_\pi] = V_t$

$i_b = \frac{V_t}{R_s \| R_1 \| R_3 + r_\pi}$

$R_o = \frac{V_t}{\frac{V_t}{R_4} + (\beta+1) \cdot \frac{V_t}{R_s \| R_1 \| R_3 + r_\pi}}$

$= \frac{V_t}{\frac{V_t}{R_4} + \frac{V_t}{\frac{R_s \| R_1 \| R_3 + r_\pi}{(\beta+1)}}} = R_4 \| \left[\frac{R_s \| R_1 \| R_3 + r_\pi}{\beta+1} \right]$